5.17 A current distribution $\vec{J}(\vec{x}\,)$ exists in a medium of unit relative permeability adjacent to a semi-infinite slab of material having relative permeability μ_r and filling the half-space, $z < 0$.

a) Show that for $z > 0$ the magnetic induction can be calculated by replacing the medium of permeability μ_r by an image current distribution, $\vec{J}^*$, with components,

$$\left(\frac{\mu_r - 1}{\mu_r + 1}\right) J_x(x, y, -z), \qquad \left(\frac{\mu_r - 1}{\mu_r + 1}\right) J_y(x, y, -z), \qquad -\left(\frac{\mu_r - 1}{\mu_r + 1}\right) J_z(x, y, -z)$$

We will end up solving parts *a*) and *b*) simultaneously. We start, however, by defining the reflection (Parity) operator $P : z \to -z$ so that

$$P : (x, y, z) \to (x, y, -z)$$

On the right ($z > 0$), we assume the magnetic induction is generated by both the original current $\vec{J}$ (contained entirely on the right) and an image current $\vec{J}^*$ (contained entirely on the left). Thus

$$\vec{B}_R(\vec{x}\,) = \frac{\mu_0}{4\pi} \int \frac{(\vec{J}(\vec{x}\,') + \vec{J}^*(\vec{x}\,')) \times (\vec{x} - \vec{x}\,')}{|\vec{x} - \vec{x}\,'|^3}\, d^3x'$$

By changing variables $z' \to -z'$ in the $\vec{J}^*$ term, we may restrict this volume integral to $z' > 0$

$$\vec{B}_R(\vec{x}\,) = \frac{\mu_0}{4\pi} \int_{z'>0} \left(\frac{\vec{J}(\vec{x}\,') \times (\vec{x} - \vec{x}\,')}{|\vec{x} - \vec{x}\,'|^3} + \frac{\vec{J}^*(P\vec{x}\,') \times (\vec{x} - P\vec{x}\,')}{|\vec{x} - P\vec{x}\,'|^3} \right) d^3x' \qquad (3)$$

On the left ($z < 0$), we assume the magnetic induction is generated by a current of the same form as the original $\vec{J}$, but with possibly modified strength (because of the change of permeability). Given a modified current $\lambda\vec{J}$ and permeability μ, we write

$$\vec{B}_L(\vec{x}) = \frac{\mu\lambda}{4\pi}\int_{z'>0} \frac{\vec{J}(\vec{x}\,') \times (\vec{x} - \vec{x}\,')}{|\vec{x} - \vec{x}\,'|^3} d^3x' \tag{4}$$

Our aim is now to match the left and right magnetic field and magnetic induction. More precisely, at $z = 0$, both H_x and H_y (the parallel components) must be continuous, and B_z (the perpendicular component) must also be continuous. To perform this matching, we first note that the norms $|\vec{x} - \vec{x}\,'|$ and $|\vec{x} - P\vec{x}\,'|$ are identical at $z = 0$. (The are both equal to $\sqrt{(x-x')^2 + (y-y')^2 + z'^2}$.) Thus all denominators are the same, and we deduce that the numerators of (3) and (4) must be matched as appropriate. For B_z, we have

$$(J_x + J_x^*)(y - y') - (J_y + J_y^*)(x - x') = \mu_r\lambda(J_x(y - y') - J_y(x - x'))$$

where any component of $\vec{J}^*$ is understood to have argument $P\vec{x}$. For H_x and H_y matching, we find

$$\begin{aligned} -(J_y - J_y^*)z' - (J_z + J_z^*)(x - x') &= \lambda(-J_y z' - J_z(x - x')) \\ (J_z + J_z^*)(x - x') + (J_x - J_x^*)z' &= \lambda(J_z(x - x') + J_x z') \end{aligned}$$

Since these equations hold for all values of (x, y), they separate into

$$\begin{aligned} \lambda J_y &= J_y - J_y^* & \qquad \lambda J_z &= J_z + J_z^* \\ \lambda J_z &= J_z + J_z^* & \qquad \lambda J_x &= J_x - J_x^* \\ \mu_r\lambda J_x &= J_x + J_x^* & \qquad \mu_r\lambda J_y &= J_y + J_y^* \end{aligned}$$

These equations may be solved to yield

$$J_x^* = (1 - \lambda)J_x, \qquad J_y^* = (1 - \lambda)J_y, \qquad J_z = -(1 - \lambda)J_z$$

provided $\mu_r\lambda - 1 = 1 - \lambda$, or $\lambda = 2/(\mu_r + 1)$. This may be given in a more concise form using the reflection operator

$$\vec{J}^*(\vec{x}) = (1 - \lambda)P\vec{J}(P\vec{x}) = \frac{\mu_r - 1}{\mu_r + 1}P\vec{J}(P\vec{x})$$

b) Show that for $z < 0$ the magnetic induction appears to be due to a current distribution $[2\mu_r/(\mu_r + 1)]\vec{J}$ in a medium of unit relative permeability.

From the expression (4) for $\vec{B}_L$, the magnetic induction appears to be due to a current $\lambda\vec{J} = [2/(\mu_r + 1)]\vec{J}$ in a medium of permeability μ. This is equivalent

to having a current distribution $[2\mu_r/(\mu_r+1)]\vec{J}$ in a medium of *unit* relative permeability.

5.19 A magnetically "hard" material is in the shape of a right circular cylinder of length L and radius a. The cylinder has a permanent magnetization M_0, uniform throughout its volume and parallel to its axis.

a) Determine the magnetic field $\vec{H}$ and magnetic induction $\vec{B}$ at all points on the axis of the cylinder, both inside and outside.

We use a magnetic scalar potential and the expression

$$\Phi_M = -\frac{1}{4\pi}\int_V \frac{\vec{\nabla}\cdot\vec{M}(\vec{x}\,')}{|\vec{x}-\vec{x}\,'|}\,d^3x' + \frac{1}{4\pi}\oint_S \frac{\hat{n}'\cdot\vec{M}(\vec{x}\,')}{|\vec{x}-\vec{x}\,'|}\,da'$$

Orienting the cylinder along the z axis, we take a uniform magnetization $\vec{M} = M_0\hat{z}$. In this case the volume integral drops out, and the surface integral only picks up contributions on the endcaps. Thus

$$\Phi_M = \frac{M_0}{4\pi}\left[\int_{\text{top}} \frac{1}{|\vec{x}-\vec{x}\,'|}\,da' - \int_{\text{bottom}} \frac{1}{|\vec{x}-\vec{x}\,'|}\,da'\right]$$

where 'top' and 'bottom' denote $z = \pm L/2$, and the integrals are restricted to $\rho < a$. On axis ($\rho = 0$) we have simply

$$\begin{aligned}
\Phi_M(z) &= \frac{M_0}{4\pi}\int\left(\frac{1}{\sqrt{\rho^2+(z-L/2)^2}} - \frac{1}{\sqrt{\rho^2+(z+L/2)^2}}\right)\rho\,d\rho\,d\phi \\
&= \frac{M_0}{4}\int_0^{a^2}\left(\frac{1}{\sqrt{\rho^2+(z-L/2)^2}} - \frac{1}{\sqrt{\rho^2+(z+L/2)^2}}\right)d\rho^2 \\
&= \frac{M_0}{2}\left[\sqrt{a^2+(z-L/2)^2} - \sqrt{a^2+(z+L/2)^2} - |z-L/2| + |z+L/2|\right]
\end{aligned}$$

On axis, the field can only point in the z direction. It is given by

$$\begin{aligned}
H_z = -\partial_z\Phi_M = -\frac{M_0}{2}\Bigg[&\frac{z-L/2}{\sqrt{a^2+(z-L/2)^2}} - \frac{z+L/2}{\sqrt{a^2+(z+L/2)^2}} \\
&- \operatorname{sgn}(z-L/2) + \operatorname{sgn}(z+L/2)\Bigg]
\end{aligned}$$

Note that the last two terms cancel when $|z| > L/2$, but add up to 2 inside the magnet. Thus we may write

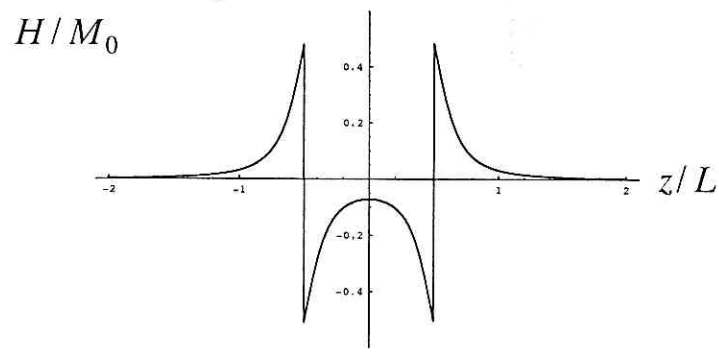
$$H_z = -\frac{M_0}{2}\left[\frac{z-L/2}{\sqrt{a^2+(z-L/2)^2}} - \frac{z+L/2}{\sqrt{a^2+(z+L/2)^2}} + 2\,\Theta(L/2-|z|)\right]$$

where $\Theta(\xi)$ denotes the unit step function, $\Theta = 1$ for $\xi > 0$ (and 0 otherwise). The magnetic induction is obtained by rewriting the relation $\vec{H} = \vec{B}/\mu_0 - \vec{M}$ as $\vec{B} = \mu_0(\vec{H} + \vec{M})$. Since the magnetization is only nonzero inside the magnet [ie $M_z = M_0\,\Theta(L/2 - |z|)$], the addition $\vec{H} + \vec{M}$ simply removes the step function term. We find

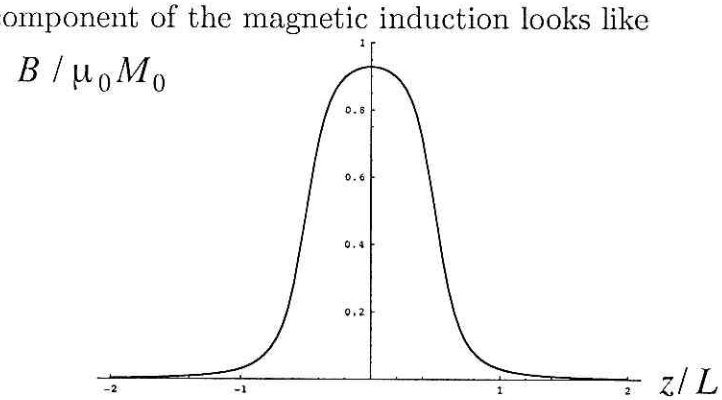
$$B_z = \mu_0(H_z + M_z) = -\frac{\mu_0 M_0}{2}\left[\frac{z - L/2}{\sqrt{a^2 + (z - L/2)^2}} - \frac{z + L/2}{\sqrt{a^2 + (z + L/2)^2}}\right]$$

b) Plot the ratios $\vec{B}/\mu_0 M_0$ and $\vec{H}/M_0$ on the axis as functions of z for $L/a = 5$.

The z component of the magnetic field looks like



while the z component of the magnetic induction looks like



Note that B_z is continuous, while H_z jumps at the ends of the magnet. This jump may be thought of as arising from effective magnetic surface charge.